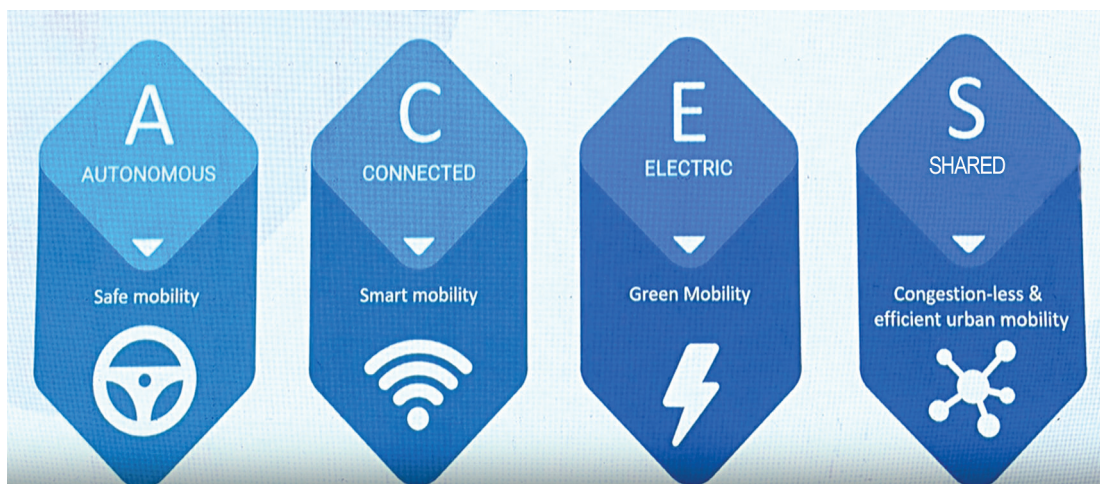


The Evolving Landscape Of VEHICLE CONTROL UNITS

Discover the latest advancements in vehicle control units, which include enhancing EV safety, integrating cutting-edge technologies, and steering towards sustainable mobility for a smarter, greener future



ASHWIN BADRI,
Chief Technology
Officer, Blaer
Motors

In the evolving landscape of control systems in the automotive industry, as we navigate towards a greener and more technologically advanced future, it is essential to understand the pivotal role that control systems play in this transformation. The automotive industry is on the brink of a transformative revolution. Control systems are at the heart of this evolution as we embrace electric vehicles, autonomous driving, and shared mobility. Traditionally, automotive control systems have been the brains of vehicles, coordinating and regulating various subsystems to ensure smooth and safe operation.

While control systems have long been a staple in four-wheelers, the landscape rapidly changes as one moves towards electric vehicles (EVs). However, the focus on control systems is not exclusive to four-wheelers; it also extends to two-wheelers.

The automotive industry is currently

guided by four pillars: autonomous driving, electrification, enhanced safety features, and shared mobility. This is depicted as follows:

Achieving these goals necessitates a seamless integration of control systems that communicate effectively with all vehicle subsystems. This synergy is critical to realising advanced driver-assistance systems, especially in reducing road accidents. Besides technological advancements, environmental concerns are a significant driving force behind the shift towards electric vehicles. Governments worldwide are pushing for reduced emissions and sustainable transportation options. To meet these objectives, adopting control systems that enhance performance and reduce greenhouse gas emissions is crucial.

The evolving control systems

While control systems in four-wheelers have seen substantial development, the focus